

# Predicting the Discrete-Log Learnability Boundary: Representation, Spectral Accessibility, and Grokking

David Vesterlund

September 2026

## Abstract

We investigate whether representation-level and architecture-conditioned spectral measurements obtained *before training* can prospectively predict whether a discrete-logarithm (DLP) formulation will directly generalize, grok after memorization, memorize without generalizing within a fixed compute horizon, or fail to fit. Our study tests two predictor classes: (A) a strict pre-training predictor using only  $t = 0$  quantities, and (B) an early-training predictor using a pre-registered 1% probe window. The primary causal representation intervention uses information-preserving GF(2) affine bijections on fixed-length binary encodings:  $r_{A,c}(z) = Az + c \bmod 2$  where  $A \in \text{GL}(B, 2)$  and  $c \in \{0, 1\}^B$ . These transforms preserve bit length, alphabet, information content, and task labels while changing only the coordinate geometry. We position our work relative to Kumar et al. (lazy-to-rich transition), Mohamadi et al. (kernel-regime limitations), Kam et al. (algebraic representability), and Song & Ye (competing memorization/generalization timescales). The central evidentiary chain is: *define predictor*  $\rightarrow$  *freeze predictor*  $\rightarrow$  *freeze configurations*  $\rightarrow$  *predict unseen runs*  $\rightarrow$  *commit predictions*  $\rightarrow$  *train*  $\rightarrow$  *evaluate without refitting*. If the spectral predictor does not outperform simple scale/data baselines, we report that result rather than changing the metric until it does.

## 1 Introduction

The *grokking* phenomenon (Power et al., 2022)—in which a neural network abruptly transitions from memorization to generalization long after overfitting—has been studied extensively on modular arithmetic tasks. A natural question is whether grokking extends to the discrete logarithm problem (DLP), a structured inverse problem on cyclic groups.

A prior study (Vesterlund, 2026) demonstrated that Transformers can grok the variable-base finite-field DLP at  $p = 113$ , including in a prime-order subgroup of order  $q = 113$ . This result coexists with the theoretical hardness result of Takhanov et al. (2024), who proved that gradient-based methods cannot efficiently learn the parity bit of the discrete logarithm in cyclic groups of large prime order.

The missing object is not merely the *learnability boundary*—the surface separating successful generalization from failure—but the ability to *predict* it before training. This paper asks:

*Can representation-level and architecture-conditioned spectral measurements obtained before training predict whether a DLP formulation will directly generalize, grok after memorization, memorize without generalizing, or fail to fit?*

**Literature positioning.** We explicitly position relative to:

- Kumar et al. (2023): Grokking as the transition from lazy to rich training dynamics. Initial feature/NTK alignment and lazy-to-rich feature learning already exist as explanations of grokking. We do NOT claim NTK-target alignment itself is novel.
- Mohamadi et al. (2024): Kernel-regime predictors can fail on modular addition because later feature learning escapes the initial kernel regime. A static  $t = 0$  measure may fail despite later grokking.
- Kam et al. (2026): Algebraic representability as the limiting regime of grokking. Representability is a prior gate.
- Song and Ye (2025): Model capacity determines grokking through competing memorization and generalization speeds.  $T_{\text{mem}}$  and  $T_{\text{gen}}$  are distinct measurable timescales.
- Takhanov et al. (2024): Gradient-based hardness of learning the discrete-log parity concept.
- Nanda et al. (2023): Fourier/mechanistic structure in grokking modular addition.

The novelty we are testing is NOT “NTK alignment matters” and NOT “representation matters.” The intended novelty is: *whether spectral and gradient accessibility can prospectively predict learning phase and time-to-generalization across DLP formulations and information-preserving representations.*

## Contributions.

1. We design a prospective prediction protocol with strict locking: predictions must precede the training they predict. Two predictor classes (pre-training and early-probe) are tested separately.
2. We use information-preserving GF(2) affine bijections as the primary causal representation intervention: same bits, same length, same alphabet, same information, same labels, different coordinate geometry.
3. We define spectral accessibility features (Fourier entropy, kernel spectral accessibility  $T_{\text{spec}}$ ,  $\text{KSA}_\tau$ ) and gradient accessibility features measured at initialization.
4. We pre-register hypotheses and success criteria before confirmatory experiments.

## 2 Related Work

### 2.1 Grokking and Training Dynamics

Power et al. (2022) introduced grokking on modular arithmetic. Kumar et al. (2023) explained grokking as a transition from lazy to rich training dynamics, showing that initial feature/NTK alignment predicts which tasks will grok. Mohamadi et al. (2024) showed that kernel-regime predictors can fail on modular addition because later feature learning escapes the initial kernel regime, meaning a static  $t = 0$  measure may fail despite later grokking.

### 2.2 Algebraic Representability and Capacity

Kam et al. (2026) established algebraic representability as the limiting regime of grokking. Song and Ye (2025) showed that model capacity determines grokking through competing memorization and generalization speeds, with  $T_{\text{mem}}$  and  $T_{\text{gen}}$  as distinct measurable timescales.

### 2.3 DLP Hardness

Takhanov et al. (2024) proved that gradient-based methods require exponentially many steps (in  $\log q$ ) to learn the parity bit of the discrete logarithm in cyclic groups of prime order. The proof relies on gradient concentration: the expected gradient becomes exponentially small as  $q$  grows.

### 2.4 Mechanistic Interpretability

Nanda et al. (2023) analyzed Fourier/mechanistic structure in grokking modular addition, showing that networks learn Fourier features.

## 3 DLP Learning Tasks

### 3.1 Cyclic Groups

For a prime  $q$ , we consider two realizations of the cyclic group  $\mathbb{Z}_q$ :

- Additive:  $G_{\text{add}} = (\mathbb{Z}/q\mathbb{Z}, +)$
- Multiplicative:  $G_{\text{mult}} = \langle g_0 \rangle \subset \mathbb{F}_p^*$  with  $|G_{\text{mult}}| = q$ , where  $p = 2q + 1$  is a safe prime.

### 3.2 Task Types

**Variable-base full-log DLP.** Given base  $g$  and target  $h = g^x$ , predict  $x$ .

**Fixed-base parity concept.** Given fixed base  $g$  and target  $h = g^x$ , predict  $\text{bit}_i(x)$ . This corresponds to the concept class in Takhanov et al. (2024).

## 4 Representations and Information-Preserving Encodings

### 4.1 Primary Intervention: GF(2) Affine Bijections

The main causal representation experiment uses information-preserving affine bijections on fixed-length binary representations:

$$r_{A,c}(z) = Az + c \bmod 2 \tag{1}$$

where  $A \in \text{GL}(B, 2)$  (invertible  $B \times B$  matrix over GF(2)) and  $c \in \{0, 1\}^B$ .

This transform preserves:

- Bit length (sequence length unchanged)
- Alphabet (binary  $\{0, 1\}$ )
- Information content (bijection)
- Task labels (labels depend on latent  $x$ , not on encoding)
- Latent examples (same manifest)

Only the coordinate geometry changes. This is a cleaner causal intervention than comparing additive vs. multiplicative groups, which changes too many things simultaneously.

## 4.2 Secondary: Additive vs. Multiplicative

Retained as a motivating secondary demonstration. Same latent  $(a, x)$  pairs, different group realizations. Not the primary causal test.

## 4.3 Encoding Families

- Identity binary:  $r(z) = z$
- Gray code: standard binary-to-Gray mapping
- GF(2) affine:  $r(z) = Az + c \bmod 2$  (primary intervention)
- Feistel: keyed nonlinear bijection (for OOD holdout)
- Atomic integer: one token per residue (secondary, vocabulary differs)

# 5 Spectral Accessibility

## 5.1 Fourier Features

For the target function on a cyclic group of order  $q$ , we compute:

- Fourier entropy  $H_F$
- Effective support  $\exp(H_F)$
- Participation ratio  $\text{PR}_F$
- Top- $k$  energy concentration ( $k \in \{1, 4, 8, 16\}$ )
- Spectral effective rank

## 5.2 Kernel Spectral Accessibility

Let  $K = U\Lambda U^T$  be the kernel eigendecomposition and  $C_j = U_j^T Y$  the target projection onto kernel eigenvectors. Under fixed-kernel gradient flow, the normalized residual energy is:

$$E(s) = \frac{\sum_j \|C_j\|^2 \exp(-2\lambda_j s)}{\sum_j \|C_j\|^2} \quad (2)$$

The *spectral accessibility time* is:

$$T_{\text{spec}}(\epsilon) = \inf\{s : E(s) \leq \epsilon\} \quad (3)$$

with primary  $\epsilon = 0.10$ .

The *Kernel Spectral Accessibility* score is:

$$\text{KSA}_\tau = \frac{\sum_j \frac{\lambda_j}{\lambda_j + \tau} \|C_j\|^2}{\|Y\|^2} \quad (4)$$

with primary scale  $\tau_0 = \text{tr}(K)/n$ .

## 6 Fixed-Kernel Theory

For squared loss under linearized gradient flow:

$$r(t) = \exp(-Kt) r(0) \tag{5}$$

and therefore spectral residual energy evolves according to kernel eigenvalues. We derive a proposition connecting target spectral mass, kernel eigenvalues, and  $\epsilon$  hitting time to  $T_{\text{spec}}$ .

We clearly state:

- Exact assumptions (squared loss, linearized dynamics)
- Whether results apply to gradient flow (yes) vs. Adam/AdamW (approximate)
- Why empirical validation is still required

## 7 Prospective Prediction Protocol

### 7.1 Two Predictor Classes

**P0: Strict Pre-Training.** Uses only  $t = 0$  quantities:  $q$ ,  $N$ ,  $P$ , target type, optimizer config, representation descriptors, Fourier descriptors, kernel alignment,  $T_{\text{spec}}$ ,  $\text{KSA}_\tau$ , initialization gradient features. May NOT use any training results.

**P1: Early-Probe.** Runs exactly 1% of training budget, then pauses. Computes kernel/gradient drift features. Makes and locks new predictions. Then resumes remaining 99%.

### 7.2 Locking Protocol

No confirmatory run may start without a valid prediction lock. Locks are append-only and contain SHA256 hashes of all relevant files. The launcher refuses to start confirmatory jobs unless `verify_lock` passes.

### 7.3 No Peeking Rule

During confirmatory phases:

- Do not inspect test curves before predictions are locked
- Do not change  $\tau$ ,  $\epsilon$ , feature sets, phase thresholds, or model coefficients
- Do not change selected encodings

### 7.4 Study Layers

Every run belongs to exactly one evidence class:

- **LEGACY\_EXPLORATORY:** pre-protocol runs, calibration only
- **CALIBRATION:** used to develop and fit the predictor
- **CONFIRMATORY:** used for blind prospective evaluation

## 8 Calibration Experiments

### 8.1 CAL\_A: Clean q=113 Pilot

40 runs: 4 encodings (identity, Gray, GF2 seed A, GF2 seed B)  $\times$  2 optimization regimes  $\times$  5 seeds. Evidence class: CALIBRATION.

### 8.2 CAL\_B: Boundary Discovery Grid

216 runs: 3 groups  $\times$  2 tasks  $\times$  4 encodings  $\times$  3 train fractions  $\times$  3 seeds. Purpose: discover phase transitions, train initial predictor.

### 8.3 CAL\_C: Capacity Sweep

120 runs: 6 boundary configurations  $\times$  4 capacities  $\times$  5 seeds. Estimate  $T_{\text{mem}}(P)$  and  $T_{\text{gen}}(P)$ .

### 8.4 CAL\_D: Optimization Sweep

120 runs: 6 configurations  $\times$  4 optimization conditions  $\times$  5 seeds.

### 8.5 CAL\_R: Random-Label Controls

72 runs: 6 configurations  $\times$  4 capacities  $\times$  3 seeds. Estimate memorization timescale separately from algorithmic generalization.

## 9 Blind Confirmatory Experiments

### 9.1 CONF\_A: Prospective GF(2) Encoding Prediction

270 runs: 3 held-out group orders  $\times$  18 encodings (6 high + 6 mid + 6 low predicted accessibility)  $\times$  5 seeds. P0 lock before training, P1 lock at 1%.

### 9.2 CONF\_B: Design/Steering Experiment

100 runs: 4th held-out  $q$ , search  $\text{GF}(B,2)$  space for high/low accessibility encodings. Tests whether predictor can construct representations that systematically shift  $T_{\text{gen}}$ .

### 9.3 CONF\_C: Out-of-Family Encoding Holdout

120 runs: 2 held-out  $q$  values  $\times$  6 encodings (identity + 5 Feistel)  $\times$  2 train fractions  $\times$  5 seeds. Tests transfer beyond linear  $\text{GF}(2)$ .

### 9.4 CONF\_D: Isomorphic Group Demonstration

20 runs: 2 group sizes  $\times$  2 realizations (additive/multiplicative)  $\times$  5 seeds. Secondary illustration.

## 10 Outcome Definitions

Using sustained thresholds ( $K = 10$  consecutive evaluations):

$$T_{\text{mem}} = \min\{t : A_{\text{train}}(t) \geq 0.99 \text{ for } K \text{ evals}\} \quad (6)$$

$$T_{\text{gen}} = \min\{t : A_{\text{test}}(t) \geq 0.90 \text{ for } K \text{ evals}\} \quad (7)$$

Phase taxonomy:

**Direct generalization:**  $T_{\text{gen}} \leq T_{\text{mem}}$

**Grokking:**  $T_{\text{mem}} < T_{\text{gen}} \leq T_{\text{budget}}$

**Memorized (censored):**  $T_{\text{mem}}$  exists,  $T_{\text{gen}}$  not observed before  $T_{\text{budget}}$

**Underfit:**  $T_{\text{mem}}$  not observed before  $T_{\text{budget}}$

**Partial:** intermediate sustained generalization  $< 0.90$

## 11 Statistical Analysis

### 11.1 Primary Endpoint

Prospective prediction of censored  $T_{\text{gen}}$  using survival/time-to-event metrics. Primary comparison: FULL P0 vs. Baseline 2. Success criterion:  $\Delta$  concordance-index  $\geq 0.05$  with 95% cluster-bootstrap CI excluding zero.

### 11.2 Grouped Cross-Validation

Configuration is the statistical unit. GroupKFold by  $q$  and encoding family. Seeds from one configuration remain in the same fold.

## 12 Pre-Registered Hypotheses

**H1:** Pre-training spectral accessibility predicts time-to-generalization after controlling for  $q$ ,  $N$ , model size, target type, and optimization.

**H2:** Information-preserving GF(2) bijections of the same DLP input cause systematic differences in learning dynamics, partly explained by spectral accessibility.

**H3:** On unseen group orders, encodings ranked high-accessibility before training generalize faster than low-accessibility.

**H4:** A locked predictor can construct high-accessibility encodings that systematically improve  $T_{\text{gen}}$ .

**H5:** If  $t = 0$  kernel information is insufficient, early lazy-to-rich movement during the 1% probe provides additional prospective predictive information.

## 13 Success Decision Tree

**Case A:** P0 beats Baseline 2  $\Rightarrow$  “Pre-training spectral accessibility prospectively predicts DLP learnability dynamics.”

**Case B:** P0 fails, P1 succeeds  $\Rightarrow$  “Early kernel/feature dynamics prospectively predict DLP learnability dynamics.” (Not called pre-training prediction.)

**Case C:** Neither improves over baselines  $\Rightarrow$  Report negative result: “Initial kernel accessibility does not contain sufficient information to predict rich-regime DLP generalization.”

## 14 Limitations

- Our experiments use small primes ( $q \leq 2^{18}$ ). The hardness theorem is asymptotic.
- The predictor is calibrated on a limited set of groups and encodings.
- GF(2) affine bijections are linear; nonlinear encodings (Feistel) may behave differently.
- The kernel spectral accessibility is a fixed-kernel proxy; Adam/AdamW dynamics differ.
- No experiment at small  $q$  constitutes a practical DLP attack.

## 15 Discussion

This study is designed to produce an interesting paper under multiple outcomes. If the  $t = 0$  predictor works, we have a strong result: the representation’s initial spectral geometry contains information about future grokking. If  $t = 0$  fails but the 1% probe works, we have empirical evidence that the lazy-to-rich transition, rather than initial kernel alignment, is what becomes predictive. If both fail, we have a substantial negative result: the generalizing DLP solution is so much a product of later feature learning that neither initial kernel geometry nor very early changes suffice to predict it.

## 16 Conclusion

We have designed a prospective prediction study with strict locking protocols to determine whether spectral accessibility can predict DLP learnability. The primary causal intervention uses GF(2) affine bijections—information-preserving transforms that change only coordinate geometry. Results from confirmatory experiments will determine which (if any) of the pre-registered hypotheses are supported.

## Reproducibility

All code, data manifests, prediction locks, and results are available at <https://github.com/VesterlundCoder/dlp-grokking-boundary>. The prediction protocol ensures that predictions precede the training they predict.

## Acknowledgments

Compute resources provided by LUMI-G (EuroHPC JU), project `project_465003364`.

## References

- Cadet Kam et al. Algebraic representability as the limiting regime of grokking. *arXiv preprint arXiv:2607.13749*, 2026.
- Vimal Kumar, Behrad Mohamadi, Jisoo Suh, and Vudtiwat Tan. Grokking as the transition from lazy to rich training dynamics. *arXiv preprint arXiv:2310.06110*, 2023.
- Behrad Mohamadi, Vimal Kumar, and Vudtiwat Tan. Why do you grok? *arXiv preprint arXiv:2407.12332*, 2024.
- Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. *ICLR*, 2023.
- Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. *arXiv preprint arXiv:2201.02177*, 2022.
- Rui Song and Yinan Ye. Model capacity determines grokking through competing memorisation and generalisation speeds. *arXiv preprint arXiv:2605.09724*, 2025.
- R. Takhanov, M. Tezekbayev, A. Pak, A. Bolatov, Z. Kadyrsizova, and Z. Assylbekov. Intractability of learning the discrete logarithm with gradient-based methods. *Proceedings of Machine Learning Research*, 222, 2024.
- David Vesterlund. Grokking the discrete logarithm: Scaling across model capacity and mechanistic analysis, 2026. URL <https://github.com/VesterlundCoder/dlp-grokking>.